BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH HONOUR

CSE3433 - SOFTWARE ARCHITECTURE

SEMESTER II SESSION 25/26

INDIVIDUAL LOGBOOK

PROJECT TITLE:
MICROSERVICE DISASTER RELIEF COORDINATION PLATFORM

NAME:
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NO MATRIC:
S76194

COURSE LECTURER
DR MOHAMAD NOR HASAN

Team Information

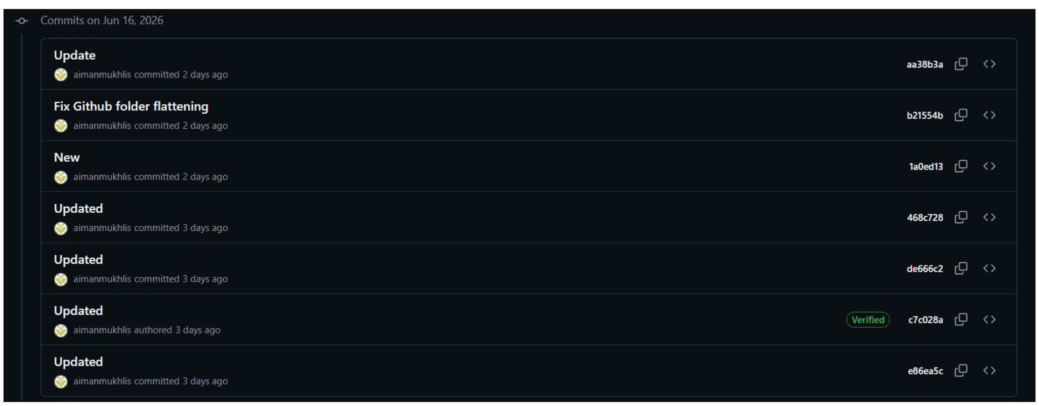
- PROJECT TITLE : MICROSERVICE DISASTER RELIEF COORDINATION PLATFORM
- GROUP : G02
- TEAM MEMBERS :
  - MUHAMMAD ADHA BIN MUHAMMAD HAMKA - PROJECT MANAGER
  - MUHAMMAD SYAKIR ZUFAYRI BIN SAARI - SYSTEM ARCHITECT

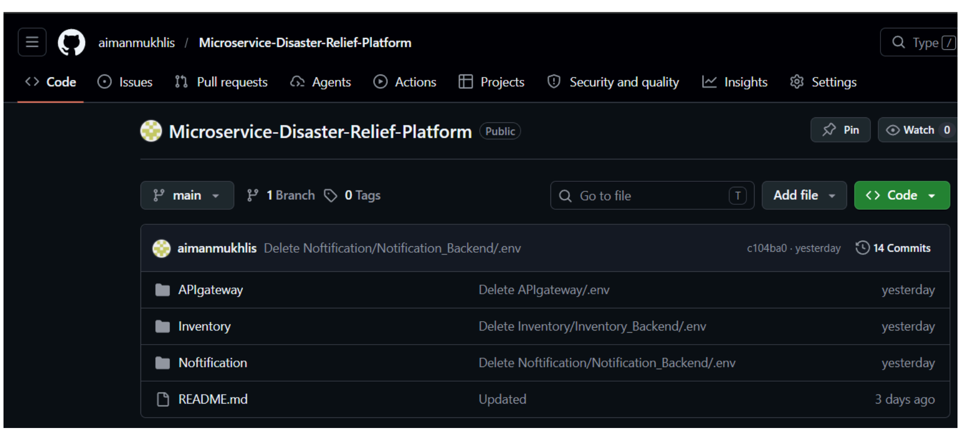
Weekly Progress

- Brainstorm Phase
    - Project Activity : Create Project Proposal Report
    - Duration : Week 4 - Week 5
    - Contribution : Suggest 4-5 Initial Functional Requirement based on Microservice Disaster Relief Coordination Platform problem description and system objective
    - Challenges :-
        - Need to analyse problem description and system objectives before proposing functional requirements.
        - Understand Microservice Architecture so the requirement can be conducted without architecture problem
    - Action Plan :-
        - Validate Initial Functional Requirement with team members
        - Analyse each FR to create module
        - Use GenAI to help in elaboration

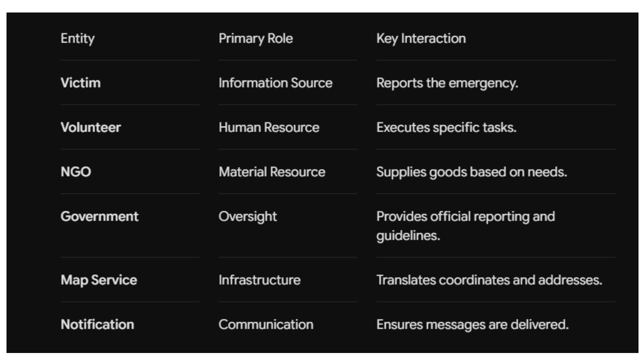
- Design Phase
    - Project Activity : Create Architecture Design Report
    - Duration : Week 6 - Week 7
    - Contribution : Improve Functional Requirement and Non-Functional Requirement based on previous initial Functional Requirement. Create System Context Diagram and Interaction Diagram (Sequence).
    - Challenges :-
        - Classify Functional Requirement according to service. So that It is loose coupling with another service
        - Need to draw sequence diagrams based on the number of services created. Exhausted way to create it 1 by 1
    - Action Plan :-
        - Break down Functional Requirement into independent microservices.
        - Use Draw.io to create interaction diagram and system context diagram with 4 major entity who use the system (NGO,Agency,Volunteer and Victim)
        - Align System Context Diagram and Interaction Diagram with 2 more diagrams created by team members
        - Analyse Proposal Report to create FR and NFR

- Prototype Phase
  - Project Activity : Implement design report to prototype project
  - Duration : Week 8 - Week 13
  - Contribution : Create Inventory service and Notification service
  - Challenge :-
    - First time develop backend, don't have knowledge to develop system according to the architecture
    - It is unclear what tools to use for backend, frontend and database
    - What is docker for?
  - Action Plan:-
    - Use AI to teach how to implement microservice disaster relief coordination platform system using node.js,mongodb and html
    - Find solution for dns problem in stack overflow
    - Analyze diagram to understand how the system work
    - Implement API Gateway to improve security of the system
  - Github log & Evidence





GenAI Usage Record

<table>
<tr><th>Date</th><th>Task</th><th>Explanation</th></tr>
<tr><td>12/4/2026</td><td>Create elaboration Initial FR</td><td>Using ChatGPT and Gemini to elaborate more about suggestion of initial FR</td></tr>
<tr><td></td><td>Prompt</td><td>“Based on this elaborate more about Functional Requirement”</td></tr>
<tr><td></td><td>Reflection</td><td>It show how the FR combine and become service</td></tr>
<tr><td></td><td>Adaptation</td><td>Take FR service which suitable for the university project microservice</td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td>27/4/2026</td><td>List NFR suitable for microservice</td><td>Tools Gemini to list NFR like scalability, portability and performance</td></tr>
<tr><td></td><td>Prompt</td><td>“Based on this file,create non functional requirement suitable for the project”</td></tr>
<tr><td></td><td>Reflection</td><td>More theory for non functional requirement such as how scaling in microservices is better than monolithic architecture</td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td></td><td>Create description System Context Diagram</td><td>Use ChatGPT to create description for Context Diagram</td></tr>
<tr><td></td><td>Prompt</td><td>“Create description based on this context diagram “</td></tr>
<tr><td></td><td>Output Summary</td><td>
<table>
<tr><th>Entity</th><th>Primary Role</th><th>Key Interaction</th></tr>
<tr><td>Victim</td><td>Information Source</td><td>Reports the emergency.</td></tr>
<tr><td>Volunteer</td><td>Human Resource</td><td>Executes specific tasks.</td></tr>
<tr><td>NGO</td><td>Material Resource</td><td>Supplies goods based on needs.</td></tr>
<tr><td>Government</td><td>Oversight</td><td>Provides official reporting and guidelines.</td></tr>
<tr><td>Map Service</td><td>Infrastructure</td><td>Translates coordinates and addresses.</td></tr>
<tr><td>Notification</td><td>Communication</td><td>Ensures messages are delivered.</td></tr>
</table></td></tr>
<tr><td></td><td>Reflection</td><td>Teach how different entities interact with the system for</td></tr>
<tr><td></td><td></td><td></td></tr>
</table>

<table>
<tr><td>28/4/2026</td><td>Create Interaction Diagram</td><td>Use ChatGPT to show interaction from user to API Gateway to each service in each module. With attachment Proposal report</td></tr>
<tr><td></td><td>Prompt</td><td>“Based on file create table to show interaction between service and user for each module to create interaction diagram”</td></tr>
<tr><td></td><td>Output Summary</td><td>
<table>
<tr><th>Module / Service</th><th>User Actor</th><th>User Action</th><th>System Response / Service Interaction</th></tr>
<tr><td>User & Authentication Service</td><td>Victim, Volunteer, NGO, Government Agency</td><td>Register account</td><td>Validate user data and store account</td></tr>
<tr><td>User & Authentication Service</td><td>All Users</td><td>Login</td><td>Verify credentials and generate JWT token</td></tr>
<tr><td>API Gateway</td><td>All Users</td><td>Access any module</td><td>Validate JWT with Authentication Service and route request</td></tr>
<tr><td>Reporting Service</td><td>Victim</td><td>Submit disaster report with GPS and photos</td><td>Store report and assign priority level</td></tr>
<tr><td>Reporting Service</td><td>Volunteer</td><td>Submit disaster report</td><td>Store report and categorize incident</td></tr>
<tr><td>Reporting Service</td><td>System</td><td>Detect critical keywords</td><td>Mark report as High Priority</td></tr>
<tr><td>Reporting Service → Notification Service</td><td>System</td><td>High-priority report created</td><td>Send alert to nearby responders</td></tr>
</table>
</td></tr>
<tr><td></td><td>Reflection</td><td>Teach me the flow of interaction for the service to work</td></tr>
<tr><td></td><td>Adaptation</td><td>According to the table, I can start create diagram to put actor,service, API Gateway and Database</td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td></td><td>Create Plantuml Interaction Diagram</td><td>Use ChatGPT to create coding for plantuml then go to plantuml to generate interaction diagram as example</td></tr>
<tr><td></td><td>Prompt</td><td>“based on this (Modules) create sequence diagram in plantuml”</td></tr>
<tr><td></td><td>Output Summary</td><td>Here is a clean PlantUML sequence diagram based on your Resource Management + Logistics Management requirements:
plantuml
<pre>
@startuml
title Resource & Logistics Management Module - Sequence Diagram

actor Requestor
actor Driver
actor "NGO / Government" as NGO

participant "API Gateway" as APIG
participant "Resource Service" as RS
participant "Logistics Service" as LS
participant "GIS Service" as GIS
database "Inventory Database" as DB

== FR-RSRC-01: Resource Request & Nearby Relief Centers (50km) ==

Requestor -> APIG: Request Resource (Item Type, Location)
APIG -> RS: Forward Resource Request

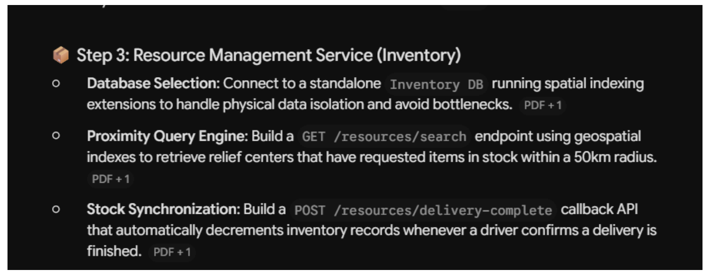
RS -> DB: Query available stock by item type
DB --> RS: Return stock data

RS -> GIS: Find relief centers within 50km radius
GIS --> RS: List of nearby centers

RS --> APIG: Return available centers with stock
APIG --> Requestor: Display resource availability

== FR6: NGO/Government Resource Registration ==

NGO -> APIG: Register Resources (Food/Medicine/Equi)
</pre>
</td></tr>
<tr><td></td><td>Reflection</td><td>I learn by generate this plantuml to figure out how to draw interaction diagram and how to improve it</td></tr>
</table>

	Adaptation	Look at generated interaction diagram as guidance
24/5/2026	Develop services backend	Ask gemini how to create inventory service backend using mongodb and node.js
	Prompt	“Show step by step how to create backend inventory service with nodejs and mongodb”
	Output Summary	 Step 3: Resource Management Service (Inventory) ◦ **Database Selection:** Connect to a standalone `Inventory DB` running spatial indexing extensions to handle physical data isolation and avoid bottlenecks. PDF + 1 ◦ **Proximity Query Engine:** Build a `GET /resources/search` endpoint using geospatial indexes to retrieve relief centers that have requested items in stock within a 50km radius. PDF + 1 ◦ **Stock Synchronization:** Build a `POST /resources/delivery-complete` callback API that automatically decrements inventory records whenever a driver confirms a delivery is finished. PDF + 1
	Reflection	Use Mongodb for standalone database and postman to testing API. To detect is it working or no
	Adaptation	Using npm command to create backend and postman to insert record into mongodb
29/5/2026	Develop service backend (notification)	Ask Gemini to create notification backend for microservice architecture in node.js
	Prompt	“How to implement notification service backend”
	Output summary	To create the **Notification Service backend** for your disaster relief platform, you need to build a system that can listen to other services (like the *Reporting* or *Communication* services) and broadcast alerts via channels like SMS, email, or mobile push notifications. Because a disaster scenario can trigger thousands of high-priority alerts simultaneously, making this service **Asynchronous (Event-Driven)** ensures it won't crash or slow down the rest of your platform. Here is a step-by-step implementation using Node.js and **RabbitMQ** as the message broker.
	Reflection	Use RabbitMQ as new tools to trigger notification service. So it will function

<table>
<tr><td></td><td>Adaptation</td><td>Use deployment diagram to connect between service</td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td>15/6/2026</td><td>Port Allocation</td><td>Using Gemini to suggest port allocation between service</td></tr>
<tr><td></td><td>Prompt</td><td>“suggest microservice port use approach Independent frontend for each service”</td></tr>
<tr><td></td><td>Output Summary</td><td>
<table>
<tr><th>Microservice Domain</th><th>Frontend UI Port</th><th>Backend Service Port</th><th>Database Por</th></tr>
<tr><td>API Gateway Layer</td><td>None</td><td>4000</td><td>None</td></tr>
<tr><td>Authentication Service</td><td>3010</td><td>3020</td><td>3306 (User D</td></tr>
<tr><td>Reporting Service</td><td>5010</td><td>5020</td><td>5432 (Repor</td></tr>
<tr><td>Resource Management Service</td><td>7010</td><td>7020</td><td>7700 (Invent</td></tr>
<tr><td>Volunteer Service (with Task Service)</td><td>9010</td><td>9020</td><td>9900 (Task D</td></tr>
<tr><td>Communication Service</td><td>6010</td><td>6020</td><td>6600 (Messa</td></tr>
<tr><td>Notification Service</td><td>None (System App)</td><td>8020</td><td>None</td></tr>
</table>
</td></tr>
<tr><td></td><td>Reflection</td><td>Each port play different role, even in 1 service have 3 port. Database port,backend port and frontend port</td></tr>
<tr><td></td><td>Adaptation</td><td>Use port to route between service</td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td></td><td>Use Docker container</td><td>Asking Gemini to teach using docker</td></tr>
<tr><td></td><td>Prompt</td><td>“how to use docker”</td></tr>
<tr><td></td><td>Output Summary</td><td>Step 1: Create a Dockerfile
Step 2: Build the Docker Image
Step 3: Run the Container</td></tr>
<tr><td></td><td>Reflection</td><td>When I deploy service on container and remove and repeat. I have remember the code to create image and run container from image</td></tr>
<tr><td></td><td>Adaptation</td><td>Using powershell to run service on container in docker desktop</td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td>16/6/2026</td><td>Use gitbash to push github repository from local</td><td>Ask Gemini to teach how to use gitbash</td></tr>
</table>

	Prompt	how to use gitbash to commit folder to the github
	Output Summary	Step 1: Open Git Bash and Navigate to Your Folder Step 2: Initialize Git in Your Folder Step 3: Stage Your Files Step 4: Commit the Files Step 5: Link Local Git to GitHub Step 6: Name Your Branch and Push!
	Reflection	Use gitbash to save time to push folder to the github
	Adaptation	Use gitbash for push and pull for faster upload and download from github